Code No: 135AE **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2024
DATA COMMUNICATION AND COMPUTER NETWORKS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is multiplexing?	[2]
b)	Classify the types of connections in data networks.	[3]
c)	What is the significance of HDLC?	[2]
d)	Discuss the role of CRC in error detection.	[3]
e)	Define network segmentation.	[2]
f)	What is logical addressing in the context of the network layer?	[3]
g)	What is data traffic in networking?	[2]
h)	Write a short note on Congestion in network communication.	[3]
i)	Define FTP and its significance in internet protocols.	[2]
j)	Differentiate between DNS and URL.	[3]

PART – B

(50 Marks)

2.a) Compare and contrast common network topologies, discussing their advantages and disadvantages.
b) Describe the ISO/OSI model in detail and explain its relevance in modern networking. [6+4]

OR

3. Explain how switching operates in different network architectures, such as circuit-switched, packet-switched, and cell-switched networks. [10]

4.a) Describe the role of the data link layer in the OSI model and its importance in networking.
b) Demonstrate the concept of channelization in the context of medium access control. [5+5]

OR

5.a) Compare and contrast point-to-point protocols commonly used in the data link layer.
b) Examine the challenges faced in data transmission over noisy channels and the techniques to mitigate them. [5+5]

6. Summarize the functioning of tunneling in networking, providing examples of its applications. [10]

OR

7.a) Determine the factors influencing the choice between unicast and multicast communication in network design.

b) Discuss the impact of network layer protocols on network performance and scalability. [5+5]

8.a) Compare and contrast the reliability mechanisms of UDP and TCP protocols.

b) Discuss the role of QoS in ensuring a seamless user experience in multimedia streaming applications. [5+5]

OR

9. Evaluate the effectiveness of differentiated services in prioritizing network traffic, citing real-world examples. [10]

10. Explain the following terms:
a) Domain Name Space
b) SMTP. [5+5]

OR

11. Outline the components and functioning of the World Wide Web (WWW), including the role of HTTP in fetching and rendering web pages. [10]

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